

Steps to complete Assignment 10 using Azure Data Factory (ADF) and related Azure services.

Step 1: Fetch All Countries Data

1. **Create Linked Services:**
 - Create a linked service to connect to the REST API.
 - Create a linked service to connect to Azure Data Lake Storage (ADLS).
2. **Create a Data Pipeline:**
 - Go to your Azure Data Factory.
 - Create a new pipeline.
 - Add a **Copy Activity** to fetch data from the REST API.
 - Configure the **Source** to use the REST API linked service with the URL `https://restcountries.com/v3.1/all`.
 - Configure the **Sink** to save the data as a Parquet file in ADLS.
3. **Publish and Trigger the Pipeline:**
 - Save and publish the pipeline.
 - Trigger the pipeline manually to ensure it works correctly.

Step 2: Fetch Data for Specific Countries

1. **Create Parameters in the Pipeline:**
 - Create parameters for the country names (e.g., `@pipeline().parameters.countryName`).
2. **Add Activities for Each Country:**
 - Create a Copy Activity for each of the 5 countries.
 - Configure the **Source** to use the REST API linked service with a dynamic URL using the parameter (e.g., `https://restcountries.com/v3.1/name/@{pipeline().parameters.countryName}`).
 - Configure the **Sink** to save each response as a JSON file in ADLS, with the filename equal to the country name (e.g., `@{pipeline().parameters.countryName}.json`).
3. **Loop Over Countries:**
 - Use a **ForEach Activity** to loop over a list of country names (`['india', 'us', 'uk', 'china', 'russia']`).
4. **Publish and Trigger the Pipeline:**
 - Save and publish the pipeline.
 - Trigger the pipeline manually to ensure it works correctly.

Step 3: Add Scheduled Triggers

1. **Create a Trigger:**
 - Go to the **Triggers** tab in Azure Data Factory.
 - Create a new trigger that runs at 12:00 AM and 12:00 PM IST.
2. **Attach Trigger to Pipeline:**
 - Attach the trigger to your pipeline.

Step 4: Add Failure Notification

- 1. **Add an Email Activity:**
 - Add an **Azure Function Activity** or **Logic App Activity** to send an email.
 - Create an Azure Function or Logic App to handle the email sending.
- 2. **Configure Failure Handling:**
 - Configure the pipeline to use the email activity as a failure handler for the Copy Activities.

Step 5: Send Email on Success and Failure

- 1. **Add Email Activities for Success:**
 - Add another email activity to send success notifications.
 - Configure the pipeline to send success emails after the Copy Activities complete successfully.
- 2. **Include Dynamic Content:**
 - Use dynamic content to include the Pipeline name, Status, SenderMailID, and Name in the email body.

Step 6: Trigger Pipeline on File Generation

- 1. **Create a Trigger for File Generation:**
 - Create a trigger based on the creation of the `India.json` file in ADLS.
- 2. **Create a Dependent Pipeline:**
 - Create a new pipeline to convert `India.json` to CSV.
 - Use a Copy Activity to copy the JSON file to a CSV file.
- 3. **Attach Trigger to Pipeline:**
 - Attach the trigger to the dependent pipeline.



